EXECUTIVE SUMMARY

UrbanCart Analytics Pipeline

Phnom Penh International University — Big Data Final Project

August 2, 2026

This report presents a comprehensive analytics pipeline built for UrbanCart, a fictional mid-size online retailer operating across six product categories and fifteen international cities. The project was designed to answer three primary leadership questions:

(1) Who are our most valuable customers and what drives their loyalty?

(2) Which products and categories are truly profitable once returns, discounts, and costs are accounted for?

(3) Is our marketing and signup data trustworthy enough to base decisions on?

Our analysis integrated data from three heterogeneous sources—a SQLite transactional database of 2,500 customers and 9,000 orders, a legacy CRM export of 1,427 customer records, and a 267-SKU supplier product catalog—into a unified, cleaned analytical dataset through a reproducible pipeline of SQL extractions, pandas-based data cleaning, and NumPy-based numerical computation.

Top Five Findings

• **Customer revenue is concentrated in at-risk, not loyal, segments.** The RFM segmentation reveals that the "At Risk" and "Lost" segments together account for $4.66 million in cumulative revenue from only 40.1% of the customer base. Champions—the most engaged, recent, and frequent buyers—contribute $1.41 million (26.3% of customers). The retention gap is UrbanCart's most urgent strategic challenge.

• **Books & Media is the revenue engine, but also a concentration risk.** With $2.80 million in net revenue and an average order value of $1,368—more than double Electronics at $599—this single category drives UrbanCart's top line. However, this dominance creates a vulnerability: any disruption to the books and media supply chain would disproportionately impact the business.

• **Data quality materially changes business conclusions.** Three extreme price outliers ($34,872, $11,507, $8,284) in the products table inflated the raw mean unit price by 41% (from $320 to $451). Without IQR-based outlier detection, any profitability-per-product or category-margin analysis would have been severely distorted—particularly for Books & Media and Beauty & Health.

• **Revenue is noisy and only mildly seasonal, with a concerning December 2024 anomaly.** Monthly revenue fluctuates ±30–40% ($327K–$515K range), with a Q3 peak and a 20.5% December 2024 month-over-month decline that contradicts retail norms. The linear time trend explains only 16% of revenue variance ($R^2 = 0.16$), meaning month-to-month factors dominate the signal.

• **Conversion is not the growth bottleneck—acquisition is.** Across all device types and countries, 92–96% of customers who browse the UrbanCart website complete at least one purchase. The platform's usability is strong; the constraint is getting more customers to the site in the first place.

Top Recommendations

• Launch a targeted re-engagement campaign for the 548 At Risk customers (combined historical value: $1.97M) before they slide into Lost status.

• Negotiate dual-supplier agreements for Books & Media to mitigate single-category revenue concentration risk.

• Institutionalize automated IQR-based outlier detection in the ETL pipeline; the 41% mean-price distortion is avoidable with basic data-quality checks.

• Investigate the December 2024 revenue drop—review inventory logs, marketing campaign calendars, and competitor activity for that period.

• Implement safety-stock-based reorder points using a 95% service level across all products, prioritizing the five highest-demand SKUs identified in the Monte Carlo simulation.